

Olist E-Commerce

Product Sales Analysis

A PostgreSQL-powered deep dive into 99,441 orders across customers, sellers, and delivery logistics

PostgreSQL

Google Data Analytics Framework

Brazilian E-Commerce

Project Overview

Background

E-commerce platforms generate massive transactional data daily. Staying competitive requires a deep understanding of:

- Customer behavior & satisfaction
- Seller performance & revenue contribution
- Product category performance
- Delivery logistics & efficiency

This project leverages PostgreSQL to transform raw transactional data into actionable business insights.

Business Questions

- 1 Which products generate the highest sales?
- 2 Which sellers contribute the most revenue?
- 3 How satisfied are customers?
- 4 How efficient is the delivery process?
- 5 How can customer retention be improved?

Dataset & Methodology

99,441

Orders

99,441

Customers

32,951

Products

3,095

Sellers

Source: Olist Brazilian E-Commerce Dataset (Kaggle)

Google Data Analytics Framework

01

Ask

Define business problems & KPIs

02

Prepare

Kaggle dataset acquisition & validation

03

Process

Cleaning, transformation & integrity checks

04

Analyze

SQL deep-dive: patterns & trends

05

Share

Looker Studio interactive dashboard

06

Act

Data-driven recommendations

Technical Stack & SQL Skills

Database Tables (9 Tables)

Customers

Orders

Order Items

Products

Sellers

Payments

Reviews

Geolocation

Product Category
Translation

SQL Techniques Applied

JOIN

Multi-table relational queries

CTE

Common Table Expressions for readability

Window Functions

RANK, ROW_NUMBER, LAG/LEAD

CASE WHEN

Conditional logic & segmentation

Aggregations

SUM, AVG, COUNT, GROUP BY

Subqueries

Nested query patterns

Data Cleaning

Handling nulls, standardizing values

Referential Integrity

Foreign key validation

Key Findings

Total Revenue

16M BRL

≈ 31.7M USD

Avg Order Value: 160.99 BRL (~\$32). Indicates healthy transaction volume and strong market presence.

Top 10 Sellers

11%

of Total Revenue

Just 10 sellers (0.3% of 3,095) drive over 11% of platform income — critical VIP partners.

Customer Retention

0%

Repeat Buyers

100% of 99,441 customers are one-time buyers. Severe retention issue requiring immediate action.

Avg Review Score

4.09

out of 5.0

Above-average satisfaction but notable friction points prevent a higher score.

Avg Delivery Time

12.5 days

Order to Delivery

A near two-week wait is the primary bottleneck capping review scores and customer satisfaction.

Actionable Recommendations

01

Implement Retention Strategy

⚠ 100% one-time buyers

Launch loyalty points, personalized email campaigns, and post-purchase vouchers to drive repeat purchases and improve Customer Lifetime Value (CLV).

03

Protect Top Sellers (VIP Program)

⚠ 0.3% of sellers = 11% revenue

Offer VIP sellers dedicated account managers, reduced commission tiers, and premium platform placement to secure their continued partnership.

02

Optimize Logistics & Delivery

⚠ 12.5-day avg delivery time

Evaluate logistics partners, add strategic fulfillment centers near high-density customer areas, and introduce premium expedited shipping options.

04

Geo-Targeted Marketing

⚠ Untapped high-AOV regions

Use geolocation data to identify cities and states with the highest Average Order Value, then focus ad spend on acquiring users in those areas.

Conclusion & Future Work

Summary

The analysis of 99,441 Olist orders uncovered significant opportunities across customer retention, seller strategy, and logistics. Revenue performance was strong at 16M BRL, but a 100% one-time buyer rate exposes a critical retention gap. The 12.5-day average delivery time is the primary driver of the 4.09/5 review score ceiling. Addressing these two core issues offers the highest potential for platform growth.

Future Improvements

- RFM Customer Segmentation
- Cohort Analysis
- PostgreSQL Data Warehouse
- Automated Reporting Pipelines

DISCLAIMER

This is a personal project created solely for educational and portfolio purposes. The data used is a public dataset obtained from Kaggle and reflects only a specific time period. This project is not affiliated with, sponsored by, or representative of the internal data or official views of any related companies.

Let's Connect

I'm open to data analyst roles, collaborations, and feedback. Feel free to reach out!



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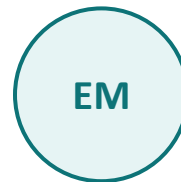
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Based in Indonesia · Open to remote & hybrid opportunities

Thank You

for taking the time to review this project.

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